

We will explain how customer demand, service time, depot information, and delivery time windows are estimated for our current 25-store Lidl case.

2. Depot and route assumptions

The following table summarizes the depot and route assumptions used for the 25-store case.

Depot parameter	Value
Depot	Dr. Oetker, Lutterstraße 14, 33617 Bielefeld
Depot node	0
Loading time	06:30–07:00
Route start	07:00
Store delivery time window	08:00–12:00
Route end	Every route returns to the depot
Shift duration	480 minutes

We assume homogeneous Euro pallets, meaning each pallet has the same size and is treated equally in the model, this simplifies the routing problem because the truck capacity can be measured only in number of pallets.

4. Demand estimation method

Exact store-level delivery quantities between Dr. Oetker and individual Lidl stores are not publicly available, therefore the demand values in this project are estimated. The values are not treated as exact real sales numbers, but as realistic input values for the routing model.

The demand is measured in pallets because the truck capacities are also given in pallets. To keep the model simple and consistent, we assume homogeneous pallets. This means that every pallet is treated as one equal loading unit in the vehicle.

Each Lidl store is classified as low, medium, or high demand. The classification is based on visible and comparable store characteristics, such as the store location, likely store size, parking or loading access, and the surrounding customer area. Smaller or more local stores receive lower demand values, while stores in stronger locations or larger catchment areas receive higher demand values.

The pallet values represent an estimated replenishment quantity for one delivery planning cycle, not exact daily product sales.

Demand class	Estimated demand	Explanation
Low	3 pallets	Smaller/local store or lower expected customer area
Medium	4 pallets	Standard Lidl store with normal customer flow
High	5 pallets	Larger store, strong location, better access or wider customer area

Since real store-level Lidl demand for Dr. Oetker products is not publicly available, the demand values are estimated. Our estimates are not random, each store is classified as low, medium, or high demand based on visible and comparable store characteristics such as store location, likely size, parking/loading access, and surrounding customer area.

Data element	Type	Source / method
Depot location	Real data	Dr. Oetker location in Bielefeld
Lidl store locations	Real data	Collected Lidl store addresses
Store demand	Estimated	Low / medium / high demand classification
Pallet type	Assumption	Homogeneous pallets
Service time	Formula-based	10 minutes base time + 2 minutes per pallet
Delivery time window	Assumption	08:00–12:00
Shift duration	Operational assumption	480 minutes
Traffic / delay buffer	Scenario assumption	Can be tested later

5. 25-store demand table

Store	Location	Justification	Demand class	Estimated demand
Lidl 1	Teutoburger Straße 98, Bielefeld	Central-east Bielefeld, urban store	Medium	4
Lidl 2	Jöllenbecker Straße 41, Bielefeld	Central/north-west, urban area	Medium	4
Lidl 3	Heeper Straße 113, Bielefeld	Central-east, good city access	Medium	4
Lidl 4	Stadttheider Straße 11, Bielefeld	North/central, strong urban catchment	High	5
Lidl 5	Detmolder Straße 345, Bielefeld	Stieghorst, district store	Medium	4

Store	Location	Justification	Demand class	Estimated demand
Lidl 6	Babenhauser Straße 18, Bielefeld	Schildesche/north-west, residential area	Medium	4
Lidl 7	Kimbernstraße 1, Bielefeld	Brackwede, district location	High	5
Lidl 8	Oldentruper Straße 255, Bielefeld	Heepen/Oldentrup, wider district reach	Medium	4
Lidl 9	Detmolder Straße 550, Bielefeld	Hillegossen/east, district edge	Medium	4
Lidl 10	Kasseler Straße 1, Bielefeld	Brackwede/Ummeln, south-west	High	5
Lidl 11	Henleinstraße 1, Bielefeld	Sennestadt, southern catchment area	High	5
Lidl 12	Neuenkirchener Str. 100, Gütersloh	Gütersloh urban district	Medium	4
Lidl 13	Carl-Bertelsmann-Straße 152, Gütersloh	Gütersloh city area, strong accessibility	High	5
Lidl 14	Auf dem Stempel 10, Gütersloh	Gütersloh, likely larger access area	High	5
Lidl 15	Westring 11, Verl	Smaller city / local catchment	Low	3
Lidl 16	Bielefelder Str. 55, Rheda-Wiedenbrück	Regional town, main road access	Medium	4
Lidl 17	Berliner Ring 40, Harsewinkel	Local/regional town	Low	3
Lidl 18	Lüchtenstraße 61, Schloß Holte-Stukenbrock	Local district store	Low	3
Lidl 19	Elsa-Brändström-Straße 1, Halle	Smaller city, local demand	Low	3
Lidl 20	Bahnhofstraße 23, Steinhagen	Smaller town and local customer base	Low	3

Store	Location	Justification	Demand class	Estimated demand
Lidl 21	Krentruper Str. 34, Leopoldshöhe	Smaller local catchment	Low	3
Lidl 22	Lange Str. 35–37, Spenge	Smaller town/local store	Low	3
Lidl 23	Friedrich-Petri-Straße 18–26, Lage	Regional town, wider area	Medium	4
Lidl 24	Ahmser Str. 50–54, Herford	Herford city area, larger city area and strong regional catchment	High	5
Lidl 25	Lambernweg 46, Enger	Smaller town/local catchment	Low	3

Total provisional demand:

3-pallet stores: $8 \text{ stores} \times 3 = 24$

4-pallet stores: $10 \text{ stores} \times 4 = 40$

5-pallet stores: $7 \text{ stores} \times 5 = 35$

Total = 99 pallets

Feasibility check

The total provisional demand is 99 pallets, this demand is a structured model assumption for one delivery planning cycle. It is not claimed to represent exact real Lidl demand, but it gives the routing model a clear and testable input value. The total demand also makes the vehicle-capacity check meaningful. If the group uses 33-pallet trucks, then 99 pallets would require at least three full 33-pallet truckloads $\rightarrow 99 \div 33 = 3 \text{ trucks}$

This does not mean that the final solution must use exactly three trucks. The final number of vehicles and routes should still be determined by the optimization model, because capacity is not the only restriction. Travel time, service time, delivery time windows, traffic buffers, and route duration also affect whether a route is feasible.

Demand class	Number of stores	Calculation	Total
Low	8 stores	8×3	24 pallets
Medium	10 stores	10×4	40 pallets
High	7 stores	7×5	35 pallets
Total	25 stores	—	99 pallets

The model does not count single product packages directly. It uses pallets as the planning unit because truck capacity is also measured in pallets. Therefore, the exact number of Dr. Oetker products per pallet is only used as a plausibility check, not as the main model unit.

Pallet capacity assumption

The routing model uses pallets as the demand unit, not individual product packages. This is necessary because the vehicle capacities are also given in pallets. Therefore, each store demand is expressed as an estimated number of pallet loading units.

As a standard loading unit, we assume homogeneous Euro pallets. The EPAL Euro pallet has dimensions of 800 mm × 1,200 mm × 144 mm, an approximate empty weight of 25 kg, and a safe working load of 1,500 kg.(1)

The exact number of Dr. Oetker products per pallet depends strongly on the product category. Frozen pizzas, baking products, desserts, and promotional items have different package sizes and carton structures. For example, Dr. Oetker Ristorante pizzas are commonly listed around 320–340 g per unit, while Dr. Oetker Professional also lists logistics data for a Pizza Snack case with 550 pieces per carton and 16 cartons per pallet for that specific item.(2)

Because the project uses a mixed Dr. Oetker delivery assumption, we do not model every product type separately. Instead, we treat one pallet as one standardized loading unit. This means the demand values in the store table represent pallet-equivalent delivery demand, not exact single-product counts.

Product / pallet assumption	Meaning for the model
Euro pallet	Standard loading unit
Pallet dimensions	800 mm × 1,200 mm × 144 mm
Pallet weight	approx. 25 kg empty
Safe working load	approx. 1,500 kg
Product units per pallet	varies by product category
Model decision	demand is measured in pallet-equivalents

Since product-level palletization data for all Lidl-specific Dr. Oetker products is not publicly available, the model does not calculate demand from exact unit counts. Instead, we use pallet-equivalent demand classes. This keeps the model realistic enough for routing while avoiding false precision.

6. Add driver break and traffic buffer

Scenario	Shift duration	Driver break	Traffic buffer	Effective route time
Optimistic	480 min	45 min	15 min	420 min
Normal	480 min	45 min	30 min	405 min
Conservative	480 min	45 min	45 min	390 min

The full driver shift is 480 minutes.

For a more realistic route check we reserve time for a break and possible traffic delay. therefore the effective route time is lower than 480 minutes and can be tested in different scenarios.

7. Service time update

our old formula:

$$s_i = 10 + 2 \times d_i$$

Still works perfectly with our new 3–5 pallet range:

Demand	Service time
3 pallets	16 min
4 pallets	18 min
5 pallets	20 min

Service time is demand-dependent because stores with more pallets need more unloading time. We use the formula $s_i = 10 + 2 \times d_i$, where 10 minutes represent a fixed base time for parking, checking, and paperwork, and 2 minutes are added for each delivered pallet. With our demand classes, service time ranges from 16 to 20 minutes per store. This keeps the model close to the group's earlier 20-minute unloading assumption but makes it more realistic because it depends on demand.

8. Time window

Parameter	Value
Store delivery time window	08:00–12:00
Depot loading time	06:30–07:00
Route start	07:00
Route end	Return to depot within available route time

For the first 25-store model, all stores use the same delivery time window from 08:00 to 12:00. This matches the group's current planning decision and keeps the first version manageable. Later, the group can make the time windows more detailed if needed.

References:

(1) <https://www.epal-pallets.org/eu-en/load-carriers/epal-euro-pallet>

(2) <https://www.knuspr.de/en-DE/2133-dr-oetker-ristorante-pizza-salame>

https://www.spinneys.com/en-ae/catalogue/dr-oetker-pizza-quattro-formagio-340g_8298/

<https://www.oetker-professional.de/produkte/non-food/non-food-sonstiges/pizza-snack-case-2in1-1-99-209139>

